

Chen Yu Yang

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Education

Hong Kong University of Science and Technology (HKUST)

Sep 2023 – June 2027

- B.Sc. in Data Science and Technology
- CGA: 3.8/4.3
- Major GPA: 3.97/4.3
- **Relevant Coursework:** Honors Mathematical Analysis(A), Honors Probability(A+), Statistical Inference(A), Bayesian Statistics(A), Statistical Machine Learning(A-), Programming with C++(A+), Honors Discrete Mathematical Tools for Computer Science(A), Multivariable Calculus(A+), Linear Algebra(A), Exploring Artificial Intelligence(A)

Technical University of Munich (TUM)

Mar 2026 – Aug 2026

- Official Exchange Program
- **Relevant Coursework:** Causal inference in time series, Deep Generative models, Introduction to deep learning, Applied Regression

Undergraduate Research Experience

Random Walks and Percolation on Graphs — UROP Guided Reading Project

2023-24

- Supervisor: NITZSCHNER, Maximilian Alexander
- Conducted guided reading on random walks and percolation on graphs, covering recurrence and transience, hitting times, stationary distributions, and asymptotic behavior on different graph structures.
- Wrote a structured summary report synthesizing key definitions, theoretical results, proof ideas, and examples from the literature.

A Multimodal Wearable and Camera-Based System for Behavior Monitoring and Personalized Intervention in Children with Autism (UROP)

2025-26

- Supervisor: OUYANG, Xiaomin
- Contributed to the setup of a multimodal data collection system combining wearable IMU sensors and camera recordings.
- Collected and organized paired IMU and video recordings for multimodal behavior analysis.
- Proposed matching IMU-derived wrist motion with video-based wrist trajectories, using monocular depth estimation from Depth Anything V2 to approximate 3D motion.
- Conducted preliminary qualitative tests of Depth Anything V2 on project videos to assess the feasibility of recovering depth-aware wrist trajectories.

Agent-Guided Real-to-Sim and World-Model Evaluation— FYP

2026–present

- Supervisor: GUO, Song — Early-stage project
- Exploring feedback-driven coding agents for Franka real-to-sim reconstruction, with validation beyond the demonstrations used for calibration.
- Planned responsibilities: develop the MuJoCo/Blender pipeline and compare agent-guided calibration with manual configuration and numerical search.
- Further planned work includes simulation-derived motion conditioning and supervised adaptation for video world models, with evaluation against held-out real-robot recordings.

Awards & Honors

HKUST Dean's List for the School of Science and the School of Engineering in Fall and Spring semesters

2024-26

HKUST S.S. Chern Class Scholarship by Department of Mathematics

2024-present

HKUST University's Scholarship for Continuing Undergraduate Students

2025-present

Research Interests

- **Research Interests:** Embodied Intelligence, Generative Model, LLM reasoning and self-improvement, reinforcement learning, multimodal understanding and generative world models

Skills

- **Computer Science:** Deep Learning: CNN-based architectures, U-Nets, ResNets, and Transformers
- Familiar with Normalizing Flow, VAE, GANs, DDPM, Flow Matching
- **Mathematics and Statistics:** Probability, statistical inference, Bayesian statistics, regression, and causal inference